

Data, plots, matrices

Notation: N rows, M attributes; i, ℓ rows, j, m attributes, c classes, k components/clusters/folds; x_i row, x_{ij} entry, y_i target, $\hat{}$ predicted/estimated.

Data matrix and standardization

Purpose: make attributes comparable before distances, PCA or penalized models.

$$\text{F1} \quad X \in \mathbb{R}^{N \times M}, \quad \tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

$$\bar{x}_j = \frac{1}{N} \sum_i x_{ij}, \quad s_j^2 = \frac{1}{N-1} \sum_i (x_{ij} - \bar{x}_j)^2.$$

Symbols: $\bar{x}_j$ and s_j are the mean and sample standard deviation of column j ; $\tilde{X}$ is the standardized matrix. **Do this:** with F1, subtract the column mean, divide by the column standard deviation, then read $\tilde{x}_{ij} = 2$ as “row i is two standard deviations above average on attribute j .”

Covariance, correlation, distance

$$\hat{\Sigma} = \frac{1}{N-1} \tilde{X}^\top \tilde{X}, \quad \rho_{jk} = \frac{\text{Cov}(x_j, x_k)}{s_j s_k},$$

$$\text{F2} \quad d_2(x_i, x_\ell) = \sqrt{\sum_m (x_{im} - x_{\ell m})^2}, \quad d_1(x, z) = \sum_m |x_m - z_m|,$$

$$d_\infty(x, z) = \max_m |x_m - z_m|.$$

Means: covariance/correlation compare columns; distances compare rows. For standardized X , $\hat{\Sigma}$ is the correlation matrix. **How to read it:** in a correlation matrix the diagonal is 1; sign gives direction, $|\rho|$ gives linear strength. In a distance matrix the diagonal is 0; the smallest off-diagonal values are nearest rows. **Do this:** for F2, choose the named distance; subtract coordinates row by row; square/sum/square-root for d_2 , sum absolute differences for d_1 , or take the largest absolute coordinate difference for d_∞ .

Plots and summary objects

Object	Read it by doing this
Histogram/density	x-axis is value; high bars/curve = many observations; wide/tails = large spread/extremes.
Boxplot	median line; box = middle 50%; long whisker/points = skew or unusual values.
Scatter plot	read axes first; upward tilt = positive correlation, downward = negative; narrow cloud = large $ \rho $.
Scatter matrix	panel (j, k) compares attributes j, k ; match each panel's sign/strength to ρ_{jk} in F2.

Recipe: identify axes and units; decide whether rows or columns are being compared; use the matching rule above and, if numeric, compute the relevant value from F1 or F2.

Matching plots to numbers

Given	How to decide
Mean/median	Symmetric plot: close together. Right tail: mean usually above median; left tail: mean below median.
Large s_j	Choose variable with widest spread on its own axis, not the tallest histogram bar.
Large $ \rho $	Choose scatter panel closest to a thin line; sign is the line's slope.
Outlier row	Look for unusual value after standardizing by F1, or large row distance in F2.

Do this: when options mix plots and matrices, first match variable names/order, then match sign, then magnitude. A correlation near 0 can still have nonlinear pattern; the formula only measures linear direction.

Regression loss

$$e_i = y_i - \hat{y}_i, \quad \text{SSE} = \sum_i e_i^2,$$

$$\text{F3} \quad \text{MSE} = \frac{1}{N} \sum_i e_i^2, \quad \text{RMSE} = \sqrt{\text{MSE}}.$$

Do this: use F3: predicted error is true minus predicted; square before averaging; RMSE is in the same unit as y . Smaller test/CV RMSE means better prediction on unseen rows.

PCA, SVD, vectors**Components, scores, reconstruction**

Purpose: replace correlated attributes by orthogonal axes ordered by variance.

$$\tilde{X} = USV^\top, \quad V = [v_1, \dots, v_M],$$

$$\text{F4} \quad Z = \tilde{X}V = US, \quad z_{ik} = \tilde{x}_i^\top v_k,$$

$$z_i = \tilde{x}_i V_K, \quad \hat{\tilde{x}}_i = z_i V_K^\top.$$

Symbols: columns of V are loading vectors; v_k is PC k ; Z contains scores; s_k is singular value; V_K keeps the first K PCs. **How to read it:** large $|v_{mk}|$ means attribute m strongly defines PC k ; a positive score means the row points in the direction of positive loadings. If all signs in one PC flip, interpretation flips but distances and variance do not change. **Projection recipe:** use F4; standardize row values first if needed; take column v_k ; multiply matching entries $\tilde{x}_{im} v_{mk}$; add them to get score z_{ik} .

Explained variance

$$\lambda_k = \frac{s_k^2}{N-1}, \quad \text{EVR}_k = \frac{s_k^2}{\sum_j s_j^2},$$

$$\text{F5} \quad \text{cumEVR}(K) = \frac{\sum_{k=1}^K s_k^2}{\sum_j s_j^2}.$$

Do this: use F5: square each singular value; divide by the total squared singular values; accumulate from PC1 upward; choose the smallest K whose cumulative value reaches the required fraction.

Interpreting a PC in words

Do this: choose one PC column in V from F4; rank variables by largest $|v_{mk}|$; describe the positive side using variables with positive loadings and the negative side using variables with negative loadings. If two variables have same sign, high scores tend to have both high; opposite signs mean the PC contrasts them. **For a score:** compute z_{ik} with F4; if z_{ik} is large positive, read row i using the positive-loading variables; if large negative, read it using the negative-loading variables.

Read PCA output and plots

Object	How to solve/read
Loading column v_k	Read down column k ; largest absolute entries name variables defining PC k ; sign tells positive/negative direction.
Score plot	Each point is one row in PC coordinates from F4; nearby points have similar standardized profiles.
Loading plot	Points/arrows are attributes; same direction = similar contribution pattern; opposite direction = negative relation in PC plane.
Biplot	Project an observation along a variable arrow; far in arrow direction means high value for that variable after scaling.

Choosing K : if the task says “at least 90% variance”, use cumulative EVR in F5. If it asks for a 2D plot, use PC1 and PC2 even if cumulative EVR is below the target, then state how much variance the two axes explain.

Norm and cosine

$$\|x\|_2 = \sqrt{x^\top x}, \quad \cos(x, y) = \frac{x^\top y}{\|x\|_2 \|y\|_2},$$

F6

$$\|X\|_F^2 = \sum_{ij} x_{ij}^2 = \sum_k s_k^2.$$

Do this: for F6, multiply matching vector entries and sum for $x^\top y$; divide by both vector lengths. Cosine near 1 means same direction; Euclidean distance also changes when vector lengths change.

Probability, Bayes, densities

Conditional probability and Bayes

Purpose: turn likelihoods and priors into class probabilities after seeing x .

$$P(A|B) = \frac{P(A, B)}{P(B)},$$

F7

$$P(A) = \sum_c P(A|c)P(c),$$

$$P(y = c|x) = \frac{P(x|y = c)P(y = c)}{\sum_{c'} P(x|y = c')P(y = c')}.$$

Symbols: prior $P(y = c)$ is class frequency before x ; likelihood $P(x|y = c)$ says how likely x is inside class c ; posterior $P(y = c|x)$ is the final class probability. **Do this:** use F7; for each class compute score = likelihood $\times$ prior; add all scores; divide each score by the sum; choose the largest posterior.

Counts and Naive Bayes

$$\hat{P}(y = c|x = a) = \frac{\#\{i : y_i = c, x_i = a\}}{\#\{i : x_i = a\}},$$

F8

$$\text{score}(c) = P(y = c) \prod_{m=1}^M P(x_m|y = c),$$

$$P(y = c|x) = \frac{\text{score}(c)}{\sum_{c'} \text{score}(c')},$$

$$\log \text{score}(c) = \log P(c) + \sum_m \log P(x_m|c).$$

How to use tables: for each class row/column, pick the probability for the observed value of each attribute; multiply them with the prior as in F8; normalize scores so posteriors sum to 1. Use log-scores when products are tiny; the largest log-score gives the same class. **From counts:** if a table gives raw counts, convert each needed entry to a conditional probability by dividing by the total count in the same condition/class. For $P(x_m = a|y = c)$, denominator is the number of rows in class c , not all rows.

Gaussian density classifier

$$\mathcal{N}(x|\mu, \Sigma) = \frac{\exp[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)]}{(2\pi)^{M/2} |\Sigma|^{1/2}},$$

F9

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right],$$

$$\hat{y} = \arg \max_c \{\mathcal{N}(x|\mu_c, \Sigma_c)P(c)\}.$$

Symbols: μ is centre, Σ covariance, σ standard deviation, $|\Sigma|$ determinant. Density height is not a probability by itself. **Do this:** use F9; for each class insert x, μ_c, Σ_c or σ_c ; multiply density by prior; choose the largest class score. Curves are highest near μ ; larger variance makes a wider, lower curve. **Boundary reading:** with equal priors, choose the class with larger density at x ; with unequal priors, a class with lower density can still win if its prior is large enough.

Prediction and boundaries

Linear and ridge regression

Purpose: predict a numeric response as intercept plus weighted attributes.

$$\bar{x} = [1, x_1, \dots, x_M]^\top, \quad X_a = [\mathbf{1}, X],$$

$$\hat{y} = \bar{x}^\top w,$$

F10

$$w_{\text{LS}} = (X_a^\top X_a)^{-1} X_a^\top y,$$

$$D = \text{diag}(0, 1, \dots, 1),$$

$$w_\lambda = (X_a^\top X_a + \lambda D)^{-1} X_a^\top y.$$

F11

$$E_\lambda(w) = \sum_i (y_i - \bar{x}_i^\top w)^2 + \lambda \sum_{j=1}^M w_j^2.$$

Symbols: w_0 is intercept; w_j is the change in $\hat{y}$ per unit x_j ; D leaves the intercept unpenalized; λ controls shrinkage. **Do this:** use F10 to add the leading 1, multiply each input by its weight, and add. For λ , compute validation/CV error for each candidate and choose the smallest; larger λ in F11 pulls non-intercept weights toward 0. **Read output:** if attributes are standardized, compare $|w_j|$ directly; otherwise coefficient size also depends on measurement units. Positive w_j raises prediction when x_j rises and all other inputs are fixed.

Logistic regression

$$a = \bar{x}^\top w, \quad \hat{p} = P(y = 1|x) = \sigma(a) = \frac{1}{1 + e^{-a}},$$

F12

$$\hat{y} = \mathbb{I}[\hat{p} \geq \tau], \quad \log \frac{\hat{p}}{1 - \hat{p}} = \bar{x}^\top w.$$

How to read it: a is log-odds; positive w_j raises positive-class probability when x_j increases. At $\tau = 0.5$, classify positive when $a \geq 0$; the boundary is $\bar{x}^\top w = 0$. **Do this:** use F12; insert the intercept 1; compute a ; convert with sigmoid; compare $\hat{p}$ with the stated threshold τ and the stated positive class. **Boundary in 2D:** solve $w_0 + w_1 x_1 + w_2 x_2 = 0$ for one axis, e.g. $x_2 = -(w_0 + w_1 x_1)/w_2$. Points on one side have $a > 0$, the other side $a < 0$.

K-nearest neighbours

$$\hat{y}_{\text{reg}}(x) = \frac{1}{K} \sum_{i \in N_K(x)} y_i,$$

F13

$$\hat{y}_{\text{class}}(x) = \arg \max_c \sum_{i \in N_K(x)} \mathbb{I}[y_i = c].$$

Symbols: $N_K(x)$ is the K training rows with smallest chosen distance to x . **Do this:** compute distances with F2; sort rows by distance; keep the first K ; average their y_i for regression or count class votes for classification. Small K gives local irregular boundaries; large K smooths.

Multiclass scores and boundary plots

F14

$$p_c = \frac{e^{a_c}}{\sum_j e^{a_j}}, \quad \hat{y} = \arg \max_c p_c.$$

Do this: use F14; exponentiate every class score, divide by the sum, then choose the largest probability. Straight boundary = linear score; axis-aligned regions = tree; local cells = KNN; smooth curved boundary = nonlinear features or neural network.

Validation and output

Hold-out and cross-validation

F15

$$E_T = \frac{1}{|T|} \sum_{i \in T} L(y_i, \hat{f}(x_i)), \quad E_{\text{CV}} = \frac{1}{K} \sum_{k=1}^K E_k.$$

Symbols: T is held-out/test rows; L is the task loss; E_k is validation-fold error. **Do this:** use F15; fit preprocessing and

model only on training rows; for each candidate compute validation or inner-CV error; choose the candidate with smallest error; report held-out/outer-fold error only after selection. **Nested CV reading:** inner folds choose hyperparameters; outer folds estimate the whole selection procedure. Do not choose by outer error first; average outer errors only after each outer training set has made its own inner choice.

Confusion matrix and metrics

$$\mathbf{F16} \quad \begin{array}{c|cc} & \hat{y} = + & \hat{y} = - \\ \hline y = + & TP & FN \\ y = - & FP & TN \end{array}$$

$$\text{Acc} = \frac{TP + TN}{N}, \quad \text{Recall} = \text{TPR} = \frac{TP}{TP + FN}, \quad \mathbf{F23}$$

$$\text{FPR} = \frac{FP}{FP + TN}, \quad \text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad F_1 = \frac{2 \text{ Precision Recall}}{\text{Precision} + \text{Recall}}. \quad \mathbf{F24}$$

Do this: first name the positive class, then fill F16. Use F17: accuracy denominator = all cases; recall/TPR = actual positives; precision = predicted positives; FPR/specificity = actual negatives. Recall asks what positives were found; precision asks what predicted positives were correct. **Matrix orientation:** if the shown matrix swaps rows/columns, do not copy positions blindly. Identify each count from the words: TP = actual positive and predicted positive; FN = actual positive predicted negative; FP = actual negative predicted positive; TN = actual negative predicted negative.

ROC curves and thresholds

$$\mathbf{18} \quad \text{ROC}(\tau) = (\text{FPR}(\tau), \text{TPR}(\tau)), \quad \text{AUC} = \text{area under ROC}.$$

Do this: sort positive scores high to low. Start at (0,0) with no predicted positives. When the next row is truly positive, increase TP and TPR using F17; when truly negative, increase FP and FPR . Lowering τ moves toward (1,1); upper-left and high AUC mean better ranking. **Threshold choice:** a higher threshold predicts fewer positives, usually lower TPR and lower FPR . If false positives are expensive, prefer a point with smaller FPR ; if missed positives are expensive, prefer larger TPR .

Paired model comparison

$$\mathbf{F19} \quad \begin{array}{c|cc} & B \text{ correct} & B \text{ wrong} \\ \hline A \text{ correct} & n_{11} & n_{12} \\ A \text{ wrong} & n_{21} & n_{22} \end{array} \quad \hat{\theta} = \frac{n_{12} - n_{21}}{N}$$

$$\mathbf{F20} \quad p = 2F_{\text{Binom}(n_{12}+n_{21}, 1/2)}(\min(n_{12}, n_{21})).$$

$$z_i = L_A(i) - L_B(i),$$

$$\mathbf{F21} \quad \bar{z} = \frac{1}{N} \sum_i z_i, \quad s_z^2 = \frac{1}{N-1} \sum_i (z_i - \bar{z})^2,$$

$$\text{CI}_{1-\alpha} : \quad \bar{z} \pm t_{1-\alpha/2, N-1} \frac{s_z}{\sqrt{N}}.$$

Do this: for classifiers use F19; only disagreements n_{12}, n_{21} affect p in F20; positive $\hat{\theta}$ favours A. For regression use F21; compute one loss difference per same test row; $\bar{z} < 0$ favours A when lower loss is better; a CI containing 0 gives no demonstrated difference.

Trees, ensembles, ANN

Decision tree split

$$I_{\text{error}} = 1 - \max_c p_c, \quad I_{\text{Gini}} = 1 - \sum_c p_c^2,$$

$$I_{\text{ent}} = - \sum_c p_c \log_2 p_c, \quad \text{gain} = I(\text{parent}) - \sum_v \frac{N_v}{N} I(\text{child } v).$$

Do this: for each candidate split, count class labels in every node; compute $p_c = n_c/N_{\text{node}}$; insert p_c in the chosen impurity in F22; multiply each child impurity by N_v/N ; choose the split with largest gain. Read a tree by following thresholds from root to leaf; classification leaf returns majority/class fractions, regression leaf returns mean response. **Regression tree:** replace class fractions by response variance/SSE inside each node if the task is numeric prediction; the chosen split is the one giving the lowest weighted child error or largest error reduction.

AdaBoost

$$\epsilon_t = \sum_i q_i^{(t)} \mathbb{I}[h_t(x_i) \neq y_i], \quad \alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t},$$

$$q_i^{(t+1)} \propto q_i^{(t)} \exp(\alpha_t \mathbb{I}[h_t(x_i) \neq y_i]), \quad H(x) = \text{sign} \sum_t \alpha_t h_t(x).$$

Do this: use signed labels/classes for h_t . In F23, add current weights $q_i^{(t)}$ only for misclassified rows to get ϵ_t ; compute α_t ; multiply each row weight by the shown factor; divide all new weights by their sum so they add to 1; final class is the sign of weighted votes.

Neural-network forward pass

$$a_j = w_{0j}^{(1)} + \sum_m w_{mj}^{(1)} x_m,$$

$$h_j = g(a_j),$$

$$\hat{y} = w_0^{(2)} + \sum_j w_j^{(2)} h_j,$$

$$\text{ReLU}(a) = \max(0, a), \quad \sigma(a) = \frac{1}{1 + e^{-a}}, \quad \tanh(a) \in [-1, 1].$$

Do this: from a diagram/table, gather all incoming weights for hidden unit j ; compute a_j with F24; apply the stated activation to get h_j ; multiply hidden outputs by outgoing weights and add output bias; apply final sigmoid/threshold only if the task says classification probability/class. **Reading weights:** a large positive incoming weight makes that input increase the hidden activation; a large negative weight makes it decrease. Bias is added even when all inputs are zero.

Clustering and outliers

K-means and centroids

$$\mathbf{F25} \quad \min_{\{C_k, \mu_k\}} \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2, \quad \mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i.$$

Do this: with F25, compute squared Euclidean distance from row x_i to every centroid μ_k ; assign to the smallest distance; recompute each centroid as the coordinate-wise mean of assigned rows; repeat only if another iteration is requested. A centroid table is the mean attribute profile of each cluster. **Read clusters:** compare centroid coordinates attribute by attribute. A high standardized centroid value means that cluster is above average on that attribute; cluster labels 1, 2, ... have no inherent order.

Hierarchical clustering and dendrograms

Linkage Compute cluster distance as

Single smallest pair distance across clusters; can form chains.
Complete largest pair distance; favours compact groups.
Average average of all cross-cluster pair distances.
Centroid distance between cluster means.

Read dendrogram: leaves are rows; merge height is the distance at which groups join; a horizontal cut creates clusters by collecting branches connected below that height. **Choosing groups:** to get exactly K clusters, cut at a height that leaves K disconnected branches. A large vertical gap means a stable separation because no merges occur over that distance range.

Compare clusterings

$$\mathbf{F26} \quad \text{Rand} = \frac{f_{11} + f_{00}}{\binom{N}{2}}, \quad J = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}.$$

Do this: inspect all row pairs. f_{11} = together in both clusterings; f_{00} = apart in both; f_{10}, f_{01} = disagreement. Rand in F26 counts both agreements; Jaccard ignores shared separation.

Gaussian mixture model

$$\mathbf{F27} \quad p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k), \quad \sum_k \pi_k = 1, \\ \gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)}.$$

Do this: use F27; for row i compute one component score π_k times Gaussian density; divide each score by the sum of all component scores; responsibilities γ_{ik} sum to 1 across k and the largest gives a hard label.

KDE, local density, ARD

$$\mathbf{F28} \quad \hat{p}(x) = \frac{1}{N} \sum_i \mathcal{N}(x|x_i, \sigma^2 I), \\ \ell_{\text{LOO}}(\sigma) = \sum_i \log \left[\frac{1}{N-1} \sum_{j \neq i} \mathcal{N}(x_i|x_j, \sigma^2 I) \right]. \\ \mathbf{F29} \quad \text{dens}(x_i, K) = \left(\frac{1}{K} \sum_{x_j \in N_i} d(x_i, x_j) \right)^{-1}, \\ \text{ARD}(x_i, K) = \frac{1}{K} \sum_{x_j \in N_i} \frac{\text{dens}(x_i, K)}{\text{dens}(x_j, K)}.$$

Do this: for KDE use F28: add Gaussian bumps from training points; for LOO exclude the evaluated point $j = i$; larger σ smooths more. For local density use F29: find K nearest neighbours, average their distances, invert; for ARD compute neighbour densities too and average density ratios. Small density or small ARD is most outlying. **Which outlier score:** KDE compares to global density; KNN density compares neighbour distances; ARD compares a point to its own neighbours. ARD is useful when clusters have different densities because it asks whether the point is sparse relative to nearby points.

Associations and sparse vectors

Association rules and Apriori

$$\text{supp}(X) = \frac{\#(X)}{N}, \\ \mathbf{F30} \quad \text{conf}(X \rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)}, \\ \text{lift}(X \rightarrow Y) = \frac{\text{conf}(X \rightarrow Y)}{\text{supp}(Y)}.$$

Symbols: X antecedent, Y consequent, disjoint itemsets; $\#(X)$ is baskets containing X . **Do this:** use F30; count baskets containing X , Y , and $X \cup Y$; support is count divided by N ; confidence is the fraction of X baskets that also contain Y ; lift compares confidence to baseline support of Y . Apriori: discard candidate itemsets below minimum support, and only extend retained frequent itemsets.

Binary attributes and Jaccard

$$\mathbf{F31} \quad \begin{array}{c|cc} & b=1 & b=0 \\ \hline a=1 & f_{11} & f_{10} \\ a=0 & f_{01} & f_{00} \end{array} \\ J(a, b) = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}, \quad \text{SMC} = \frac{f_{11} + f_{00}}{M}.$$

Do this: for each attribute, count which cell of F31 the pair (a_m, b_m) falls in. Jaccard ignores shared zeros, so use it for sparse baskets/word presence; SMC counts both shared ones and shared zeros.

Text vectors and cosine

$$\mathbf{F32} \quad \text{tfidf}_{ij} = \text{tf}_{ij} \log \frac{N}{\text{df}_j}, \quad \cos(x, y) = \frac{x^\top y}{\|x\|_2 \|y\|_2}.$$

Do this: use F32; multiply term frequency by inverse document frequency, so rare terms get larger weight; compare documents with cosine from F6. Cosine near 1 means similar word-weight direction, independent of document length scale.

Choose from the given object

Given	Use this concrete action
Class probabilities/scores	Normalize scores with F7, F8, F27 or F14; choose largest.
Weights and a new row	Add intercept, multiply weights and inputs using F10, F12 or F24.
Validation table	Choose smallest validation/inner error with F15; report held-out/outer error.
Confusion/ROC	Define positive class, fill counts, then use denominators in F17.
Centroids/dendrogram	Assign nearest centroid with F25, or cut dendrogram at requested height/ K .
Density/neighbours	Use F27, F28 or F29; smallest density/ARD is most outlying.
Baskets/text vectors	Count rule supports with F30, or compare sparse vectors with F31/F32.